STATISTICS

Chapter 4 Homework

Discrete random variables and expected value

Sections 4.1-4.6 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

One distribution, several questions

1. Let X be the number of late deliveries in a day. Complete the probability distribution and explain why it is valid.

x	0	1	2	3
P(X = x)	0.2	0.4	?	0.1

2. For Problem 1 find P(X ≥ 2), P(X > 2), and P(X ≤ 1). Explain the difference between the first two events.

3. Find the expected number of late deliveries for Problem 1. Explain why a noninteger expected value is meaningful even though X is a count.

4. For Problem 1, compute E(X squared), the variance, and SD. Use variance = E(X squared) - [E(X)] squared, and √0.81 = 0.9. What are the SD units?

5. A game pays $0, $4, or $10 with probabilities 0.5, 0.4, and 0.1. It costs $3 to play. Find expected payout, expected net gain, and a fair entry fee. Does expected loss mean every play loses money?

Choose a model from its assumptions

6. Four independent packages each have probability 0.25 of arriving late. Let Y count late packages. Name the distribution and its parameters. Find P(Y = 1), P(Y ≥ 1), and E(Y).

7. Independent attempts succeed with probability 0.2. Let T count attempts through the first success. Name the model. Find P(T = 3), P(T > 3), and E(T). State whether T counts failures or all attempts.

8. A box has 5 marked and 3 unmarked cards. Select 2 uniformly without replacement. Let H count marked cards. Identify the distribution and possible values; find P(H = 2). You may use 8 choose 2 = 28 and 5 choose 2 = 10.

9. Assume arrivals follow a constant-rate Poisson process averaging 2 per hour. State the mean count in 30 minutes. Find the probability of no arrivals in 30 minutes, using $P(X = 0) = e$ to the power -mean and e to the power $-1 \approx 0.3679$.

10. Explain which earlier model applies to (a) a fixed number of independent yes/no trials, (b) trials until first success, (c) a fixed-size sample without replacement, and (d) counts in a fixed time interval under constant-rate Poisson assumptions. For each, identify what the random variable counts.

11. Why is the number of marked cards in Problem 8 not exactly binomial? Explain what happens to the next-draw success probability after a marked card is selected.